Solving Inverse Problems: Variational Regularization vs. CNNs

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Abstract

We survey several methodologies used to solve ill-posed inverse problems in image reconstruction. A central focus is the tension between theoretical guarantees and empirical performance when applying variational regularization compared to convolutional neural networks. While fully non-convex models generally achieve superior expressivity, they require careful tuning and are computationally expensive. On the other hand, designing regularizers that preserve convexity can offer success comparable to deep learning approaches while also being more tractable.

1 Inverse Problems and Variational Regularization

In imaging sciences and signal processing, we frequently encounter the challenge of recovering an unknown clean image or signal from corrupted indirect measurements. This recovery process is typically modeled via the following linear relation:

$$y = Hx + n,$$

where $x \in \mathbb{R}^d$ represents the true clean image, $y \in \mathbb{R}^m$ is the corrupted measured data, $H \in \mathbb{R}^{m \times d}$ is the forward operator (which accounts for physical acquisition processes such as blur, subsampling, or CT projection), and $n \in \mathbb{R}^m$ denotes the measurement noise or model error. Often, this inverse problem is fundamentally *ill-posed*, meaning that recovering x from y is inherently unstable or non-unique. For instance, if the forward operator H is non-invertible or ill-conditioned, then small changes in the measurement y can induce arbitrarily large and destructive variations in the final reconstruction.

A classical and highly effective method to stabilize such ill-posed inverse problems is *variational regularization* (Hertrich (2025)). Under this framework, the unknown image x is reconstructed by solving the optimization problem:

$$\widehat{x}(y) = \arg\min_x \{\mathcal{D}(Hx, y) + \alpha \mathcal{R}(x)\}$$

The data fidelity term (or loss function), $\mathcal{D}(Hx, y)$, ensures that the reconstructed signal remains consistent with the observed measurements. The regularizer, $\mathcal{R}(x)$, encodes prior structural assumptions about the clean signal, such as sparsity or spatial smoothness. Finally, the regularization parameter $\alpha > 0$ dictates the trade-off, balancing the fit to the measurements against the enforced prior structure.

This variational approach corresponds nicely with Bayesian statistics. From a Bayesian viewpoint, the unknown image is modeled as a realization of a random variable $X \sim P_X$. By applying Bayes' rule, the posterior distribution of the image given the measurements is proportional to the product of the likelihood and the prior:

$$\underbrace{p_{X|Y}(x \mid y)}_{\text{posterior}} \propto \underbrace{p_{Y|X}(y \mid x)}_{\text{likelihood}} \quad \underbrace{p_X(x)}_{\text{clean image prior}} .$$

Consequently, taking the maximum a posteriori (MAP) estimator—which minimizes the negative log-posterior—possesses a form that perfectly mirrors the variational formulation:

$$\widehat{x}_{\text{MAP}} = \arg\min_x \left\{ -\log p_{Y|X}(y \mid x) - \log p_X(x) \right\}.$$

We easily recover the variational regularization estimator $\hat{x}$ by setting $p_{Y|X}(y \mid x) \propto e^{-\mathcal{D}(Hx,y)}$ and $p_X(x) \propto e^{-\alpha \mathcal{R}(x)}$.

Now that we are equipped with a baseline mathematical understanding of variational regularization, we delve into the different modeling approaches researchers have explored. In particular, we are concerned with parameterizing a learning the regularizer, $\mathcal{R}_\theta(x)$. The following sections introduce models of $\mathcal{R}_\theta(x)$ and discuss the fundamental trade-off between interpretability and expressivity/performance. As we sift through these modeling techniques, each subsequent section will build upon the last by adding a layer of architectural complexity while simultaneously peeling away a layer of interpretability. We start with a most simple, convex model (guaranteeing well-behaved solutions) and then progressively introduce non-convex elements until we reach the highly expressive, yet opaque, deep convolutional neural networks.

2 Convex Learned Regularizers: ICNNs

The first model we turn our attention to is the Input Convex Neural Network (ICNN), a scalar-valued neural network designed so that its output is a globally convex function of its input. Mathematically, an ICNN propagates the input x through hidden layers z_i according to:

$$\begin{aligned} z_1 &= g_0(W_0^{(x)} x + b_0), \\ z_{i+1} &= g_i(W_i^{(z)} z_i + W_i^{(x)} x + b_i), \qquad i = 1, \dots, \ell - 1. \end{aligned}$$

The regularizer is defined as the output of the final layer, $\mathcal{R}_\theta(x) = z_\ell$. Convexity is strictly enforced by requiring the following architectural constraints: the activation functions g_i must be coordinate-wise convex and non-decreasing (e.g., ReLU), and the weight matrices connecting intermediate layers $W_i^{(z)}$ must be coordinate-wise non-negative. Meanwhile, the input weights $W_i^{(x)}$ remain unrestricted so that $W_i^{(x)} x + b_i$ is an affine function of x, serving as skip-connections that inject expressivity back into the model without violating the overall convexity.

While the standard ICNN is not necessarily the absolute most theoretically expressive convex architecture in existence, we choose to discuss it here because it serves as the foundational paradigm for convex deep learning (Amos et al. (2017)). It offers a highly tractable, elegantly constrained framework that is easily implemented within standard deep learning libraries. This makes the ICNN the perfect mathematical baseline to ground our understanding before we venture into more complex, non-convex architectures.

3 Input Difference-of-Convex Neural Networks

Traditional regularizers, such as total variation or the ℓ_1-norm, offer strong theoretical optimization guarantees but are inherently limited in flexibility and may fail to capture the complexity of natural data. Conversely, data-driven regularizers that use deep neural networks (DNNs) demonstrate strong empirical performance but often result in highly nonconvex formulations that lack rigorous theoretical guarantees. To bridge this gap, Zhang and Leong (2025) propose Input Difference-of-Convex Neural Networks (IDCNNs), which involves learning a regularizer parameterized as a Difference-of-Convex (DC) function. Since DC regularizers generalize weakly convex functions, they yield improved expressivity and empirical performance while retaining the structure necessary for strong convergence guarantees.

The proposed framework parameterizes the regularizer $\mathcal{R}_\theta$ as the difference of two Input Convex Neural Networks (ICNNs). Specifically:

$$\mathcal{R}_\theta(x) = \mathcal{R}_{\theta_1}(x) - \mathcal{R}_{\theta_2}(x)$$

where both $\mathcal{R}_{\theta_1}$ and $\mathcal{R}_{\theta_2}$ are strictly convex with respect to their input variables. To ensure the convexity of these individual networks, the ICNN architecture enforces non-negative weights at each layer and utilizes convex, non-decreasing activation functions, such as ReLU or Softplus.

The overall objective for the inverse problem is formulated using a standard squared ℓ_2 data-fidelity term:

$$\min_{x\in\mathbb{R}^d} F(x) := \frac{1}{2}\|Ax-y\|_2^2 + \mathcal{R}_{\theta_1}(x) - \mathcal{R}_{\theta_2}(x)$$

For theoretical convergence to hold during optimization, it is assumed that the objective function $F(x)$ is bounded from below, and that the individual regularizers $\mathcal{R}_{\theta_1}$ and $\mathcal{R}_{\theta_2}$ are continuous and convex. Furthermore, $\mathcal{R}_{\theta_1}$ is assumed to satisfy L_1-smoothness conditions (i.e., its gradient is Lipschitz continuous).

Zhang and Leong leverage the Difference-of-Convex Algorithm (DCA), a classical majorization-minimization approach. Since the data-fit functional $\mathcal{L}(x;y) = \frac{1}{2}\|Ax-y\|_2^2$ is inherently convex, the algorithm isolates the challenging concave component, $-\mathcal{R}_{\theta_2}(x)$, and linearizes it at each iteration t to form a strictly convex surrogate majorizing function. DCA proceeds iteratively as follows:

1. **Subgradient Computation:** Compute the subgradient of the second ICNN at the current image estimate x_t:
$$g_t \in \partial\mathcal{R}_{\theta_2}(x_t)$$
2. **Majorization Formulation:** Define the strictly convex majorizing function $q(x;x_t)$ by replacing the concave term with its first-order Taylor expansion:
$$q(x;x_t) := \frac{1}{2}\|Ax-y\|_2^2 + \mathcal{R}_{\theta_1}(x) - \mathcal{R}_{\theta_2}(x_t) - \langle g_t, x-x_t\rangle$$
3. **Convex Minimization:** Solve the resulting convex optimization sub-problem to find the next iterate:
$$x_{t+1} \in \arg\min_{x\in\mathbb{R}^d} q(x;x_t)$$

By design, $q(x;x_t)$ acts as a strict upper bound to the true objective function at x_t. Therefore, solving this surrogate sub-problem at each iteration guarantees that the sequence of function values $\{F(x_t)\}$ is strictly and monotonically decreasing. Under the established smoothness assumptions, any limit point of this sequence converges to a stationary critical point of the overall inverse problem and effectively bypasses the instabilities often observed in standard DNNs.

4 Weakly-Convex Ridge Regularizers

It is known that a well-chosen non-convex regularizer $\mathcal{R}$ can lead to improved performance. The caveat is thatfinding the global minimum becomes generally intractable. One remedy is the convex non-convex (CNC) framework, where the global objective is modified to be

$$\mathcal{J}(x) = \frac{1}{2}\|Hx-y\|_2^2 + \mathcal{R}_{\mathrm{CNC}}(x).$$

Here, $\mathcal{J}(x)$ is convex and $\mathcal{R}_{\mathrm{CNC}}$ is non-convex. When the problem is ill-posed, then designing the CNC model becomes difficult. Various strategies to design a weakly convex regularizer include: 1) replacing the $\|\cdot\|_1$ by a non-convex penalty function, 2) using the generalized Moreau envelope of $\mathcal{R}_{\mathrm{CNC}}$, and 3) learning the proximal operator of $\mathcal{R}_{\mathrm{CNC}}$.

For the task of image denoising, Goujon et al. (2024) proposed weakly-convex regularizers. A regularizer $\mathcal{R}$ being ρ-weakly convex means that it satisfies

$$H_{\mathcal{R}}(x) \succeq -\rho I,$$

where $H_{\mathcal{R}}$ is the Hessian of the regularizer. For the overall objective to be convex, we require that $\rho = 1$. This ensures that the proximal operator for $\mathcal{R}$ is well-defined and that the denoiser is Lipschitz continuous.

The weakly-convex regularizer $\mathcal{R}$ is

$$\mathcal{R} : x[\cdot] \mapsto \sum_{i=1}^{N_C}\sum_{\mathbf{k}\in\mathbb{R}^2} \psi_i((h_i * x)[\mathbf{k}]),$$

where $x[\cdot]$ represents a 2-d image, h_i is the impulse response of the filters, and ψ_i is the potential function with Lipschitz continuous derivatives. Equivalently,

$$\mathcal{R} : \mathbf{x} \mapsto \sum_{j=1}^{d \times N_C} \psi_j(w_j^\top \mathbf{x}),$$

where $\mathbf{x}$ is the vectorized representation of x, and w_j corresponds to shifted versions of the filter kernels. The proximal denoiser is

$$D_\theta(\mathbf{y}) = \arg\min_{\mathbf{x} \in \mathbb{R}^d} \frac{1}{2}\|\mathbf{x} - \mathbf{y}\|_2^2 + \mathcal{R}_\theta(\mathbf{x}).$$

The authors decompose the activation ψ' as the difference of two splines and enforce the convolutional filters W to have unit spectral norm. This ensures that $\mathcal{R}$ will be 1-weakly convex, the Lipschitz constant of $\nabla\mathcal{R}$ is bounded, and ρ is controlled by the spline profiles.

The proximal denoiser $D_\theta(\mathbf{y})$ is learned as follows:

1. **Spectrum Normalization:** Use the discrete Fourier transform to compute the spectrum of W.
2. **Implicit Differentiation:** Using the deep equilibrium framework, compute the Jacobian of the denoiser via implicit differentiation.
3. **Optimization:** Solve the training problem with stochastic Adam.

5 End-to-End CNNs

A natural solution to inverse problems is the use of deep-learning to solve the problem in an end-to-end fashion without explicitly enforcing any structured prior. This section focuses on the work of Zhang et al. (2017), who apply advancements in convolutional neural networks (CNNs) and deep learning training to the image denoising problem $y = x + v$, where x is the clean latent image we want to recover, y is the noisy image we observe, and v is the additive noise. Therefore, this is a special case of an inverse linear problem with $H = I$.

Usually, image denoising methods assume v is some form of gaussian noise with a specified variance level and attempt to directly predict x from y with this knowledge. The authors propose instead to learn v (of any structure) through a deep CNN and then recovering x as y - $\hat{v}$. Therefore, as an input progresses through the network, we ideally strip its latent structure completely away. The authors postulate that in general y and x are very similar, so learning a mapping between them essentially amounts to learning an identity mapping, which DNNs struggle with.

To train a residual map $R(y; \Theta) \approx v$, they use the loss function

$$l(\Theta) = \frac{1}{2N} \sum_{i=1}^{N} ||R(y_i; \Theta) - (y_i - x_i)||_F^2$$

i.e., regressing the CNN's predicted $\hat{v}$ onto the actual v. They also use a CNN architecture (DnCNN) of depth D that exploits recent progress in deep learning training dynamics in the form of batch normalization (Ioffe and Szegedy (2015)) and is specially tuned for image denoising:

- **Layer 1** Convolution + ReLU
- **Layers 2–(D−1)** Convolution + BN + ReLU
- **Layer D** Convolution to original image size

Layer 1 uses a standard convolution layer with the ReLU activation $\mathrm{ReLU}(x) = \max(0, x)$. They use 64 convolutional filters of size $(3 \times 3 \times c)$, where c is the number of channels of the image. Layer 2 uses a standard convolution layer with 64 filters of size $(3 \times 3 \times 64)$ and then, crucially, batch normalization before ReLU activation. Batch normalization stabilizes training by normalizing, shifting, and scaling pre-activation inputs to give subsequent layers more stable input distributions. For a convolutional layer, this amounts to finding the empirical mean and variance post convolution within a channel across the batch and normalizing by these values. Then, we shift and scale this channel in the batch by learned parameters to retain network expressivity. Note that we pad with zeros at every layer to retain the original image size. Layer D uses c $(3 \times 3 \times 64)$ Convolutional filters to get the original image size as our output.

Interestingly, they find that the residual learning of $\hat{v}$ and batch normalization mutually benefit each other. This is likely because as the network strips the latent structure of the image, the Gaussian noise that remains is more amenable to normalization. The proposed DnCNN architecture beats other more hand-crafted and test-time compute heavy methods like TNRD and CSF on test set PSNR. Therefore, their flexible method, which works well for different noise levels and can be adapted to other inverse problems, serves as a strong baseline and suggests end-to-end solutions are viable for some inverse problems.

6 Experiments and Results

We empirically assess the efficacy of the discussed regularization frameworks by benchmarking their performance across multiple inverse problems in imaging. We categorize the regularizers into four distinct paradigms: convex models, weakly convex models, non-convex models, and CNNs.

6.1 Experimental Setup and Evaluation Metrics

The evaluation framework is designed to test the regularizers' expressivity, generalization capacity, and domain-specific performance. A single NVIDIA RTX 4090 (24GB) GPU is utilized for all evaluations to ensure consistent computational benchmarking.

Datasets and Tasks. The models are evaluated across three distinct experimental settings:

1. **Image Denoising (Base Expressivity):** Models are trained and evaluated on removing additive Gaussian noise ($\sigma = 0.1$) using the standard BSDS500 and BSD68 datasets. This establishes a baseline for the representational capacity of each learned prior.
2. **Out-of-Distribution Generalization:** To test how well the learned priors transfer to unseen inverse problems, models trained solely on natural image denoising are evaluated "out-of-the-box" on a medical computed tomography (CT) reconstruction task using the LoDoPaB-CT dataset, without any fine-tuning.
3. **Domain-Specific Training:** Finally, to evaluate the performance ceiling of these architectures in a specialized domain, the models are trained and tested explicitly on the LoDoPaB-CT dataset.

Evaluation Metrics. Reconstruction fidelity is quantified using the peak signal-to-noise ratio (PSNR), measured in decibels (dB), where higher values indicate closer proximity to the ground truth. Computational efficiency is assessed via two metrics: *training time* (measured in hours to capture the cost of parameter optimization) and *evaluation time* (measured in seconds to evaluate the speed of the optimization algorithm, such as nmAPG, during inference).

6.2 Comparative Analysis of Regularization Frameworks

Convex and Weakly Convex Models. Convex architectures, such as the Convex Ridge Regularizer (CRR) and ICNN, operate within highly constrained parameter spaces (approximately 15k and 26k parameters, respectively). Consequently, they yield the lowest PSNRs among the learned architectures, often producing reconstructions with overly smoothed textures that fail to capture high-frequency details. However, they consistently outperform traditional, unlearned regularizers such as Total Variation (TV). Their primary advantage lies in their unmatched computational efficiency: training typically converges in under an hour, and evaluation is near-instantaneous (< 0.2 seconds). Relaxing strict convexity to formulate WCRRs provides an effective middle ground, yielding slight PSNR improvements while retaining the rapid training and inference speeds characteristic of their convex counterparts.

Non-Convex Models and Training Sensitivity. Highly parameterized non-convex models, such as the Total Deep Variation (TDV, $\sim$400k parameters) and Learned Step Regularizer (LSR, $\sim$4M parameters), demonstrate vastly superior expressivity. These models achieve the highest PSNRs among variational methods, yielding crisp, well-defined reconstructions that rival deep supervised networks. However, this high capacity introduces extreme fragility regarding the training methodology. Achieving state-of-the-art performance necessitates computationally expensive supervised bilevel training routines (e.g., BL-IFT or BL-JFB), which can take upwards of 41 hours. When

optimized using simpler, weakly supervised or unsupervised schemes—such as Adversarial Regularization (AR) or Score Matching (SM)—the reconstruction quality collapses, resulting in either artificially smoothed or heavily artifact-laden images. Furthermore, their evaluation time is notably slower, taking 1–2 seconds per image.

End-to-End Convolutional Baselines. Massively parameterized end-to-end architectures, such as DRUNet ($\sim$32.6M) and FBP+UNet ($\sim$34.5M), serve as supervised baselines. In domain-specific tasks (Experiment 3), these models dictate the performance ceiling, heavily dominating the variational methods in terms of PSNR and visual sharpness. During the generalization experiments, end-to-end models trained explicitly for CT easily outpaced variational methods attempting to adapt from denoising. Despite this dominance, these models cannot be interpreted within a variational reconstruction framework and carry a well-documented risk of introducing hallucinated structures not present in the underlying measurement data.

7 Discussion: The Expressivity-Guarantee Trade-off

A central theme emerging from these empirical evaluations is the fundamental trade-off between theoretical guarantees and representational capacity in inverse problem solving. Strictly convex models guarantee stable optimization and global convergence; however, their geometric restrictions severely limit their ability to model the complex distributions of natural images. Conversely, high-capacity non-convex models deliver state-of-the-art image quality but sacrifice stability, becoming computationally expensive and highly sensitive to their training environments.

The IDCNN Framework as a Structural Bridge. Within this landscape, the IDCNN emerges as a highly attractive compromise. By explicitly parameterizing the regularization functional as the difference between two convex functions, the IDCNN unlocks the non-convex expressiveness required for high-fidelity texture and edge reconstruction. Concurrently, it maintains a rigorous mathematical structure that allows for the application of the DCA. By linearizing the concave portion of the objective at each step, the DCA provides a convex surrogate that acts as a strict upper bound, guaranteeing monotonic decrease and stable convergence to a stationary point. Thus, the IDCNN bridges the gap between the two paradigms, offering the visual fidelity of deep non-convex networks alongside the optimization guarantees of convex frameworks.

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